

SCRATCH — LOOPS (REPEAT BLOCK)



MEET SCRATCH!

Scratch is a fun way to learn coding by snapping colourful blocks together!

You can make characters move, dance, and talk — all by dragging and dropping blocks.

No typing needed!





**A LOOP REPEATS
ACTIONS AGAIN AND
AGAIN!
LIKE BRUSHING YOUR
TEETH EVERY
MORNING!**

WHY USE LOOPS?

Save time and effort
Make coding easier

Repeat fun actions
Help make games and animations
better



THE REPEAT BLOCK

WHAT IS THE REPEAT BLOCK?

The Repeat block tells the computer to do something many times! You pick a number — and the computer does the action that many times. No need to write it again and again!



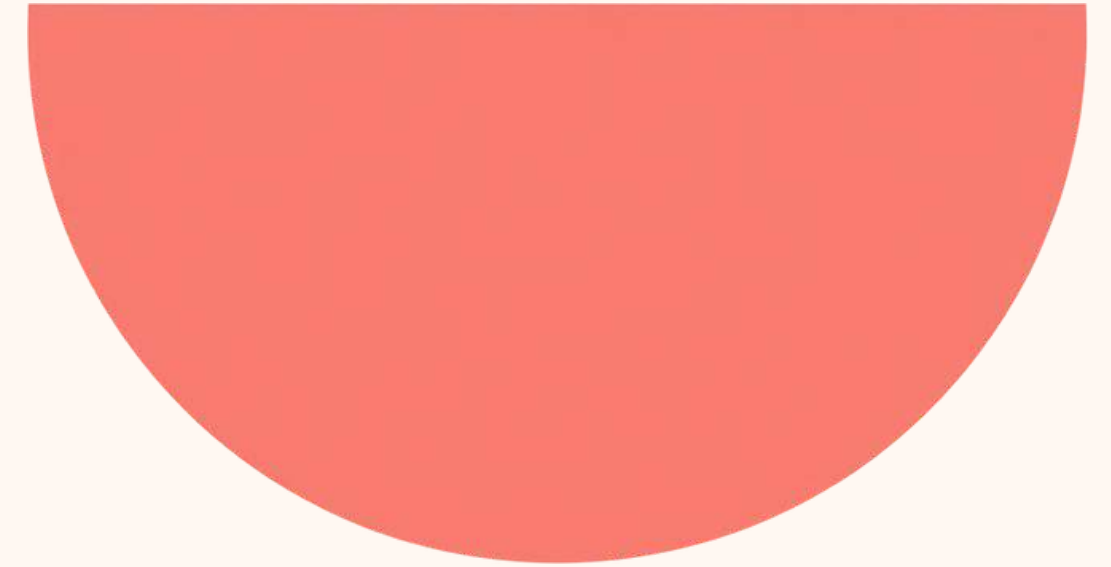
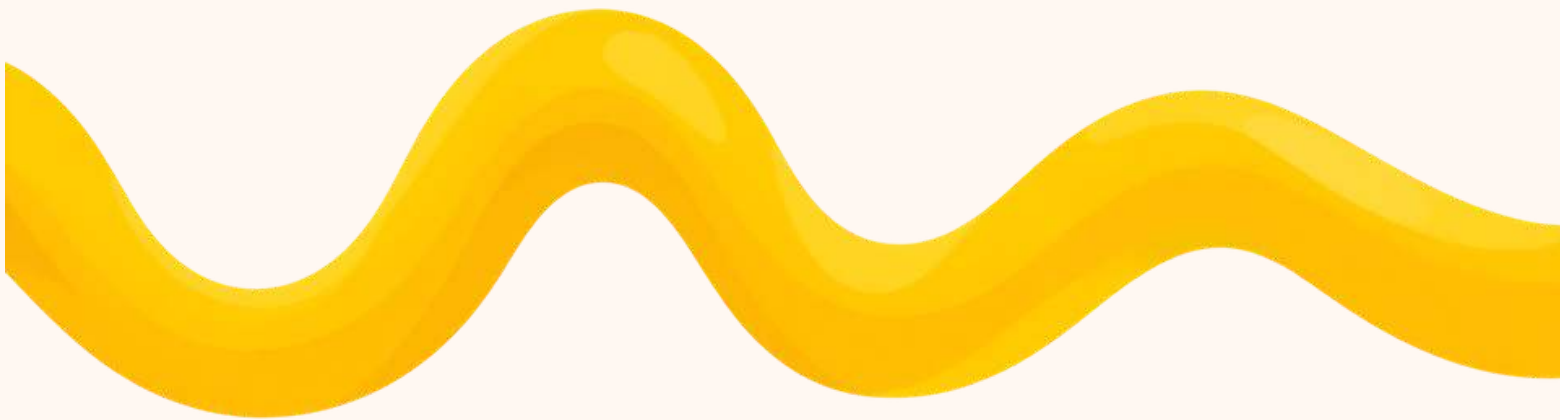
PARTS OF THE REPEAT BLOCK

-  Number input — tells how many times to repeat.
-  Blocks inside — the actions that get repeated!

HOW DOES THE REPEAT BLOCK WORK?

You pick a number — that's how many times the block repeats!

The computer does that action that many times and then stops.



LET'S TRY IT!

Example:

Repeat 4 times → the action happens 4 times!

Repeat 1 time → 1 time.

Repeat 10 times → 10 times!

You are in control! 🎉

HOW DOES THE REPEAT BLOCK WORK? (PART 2)

The computer does the instructions inside the Repeat block again and again until the number is done.

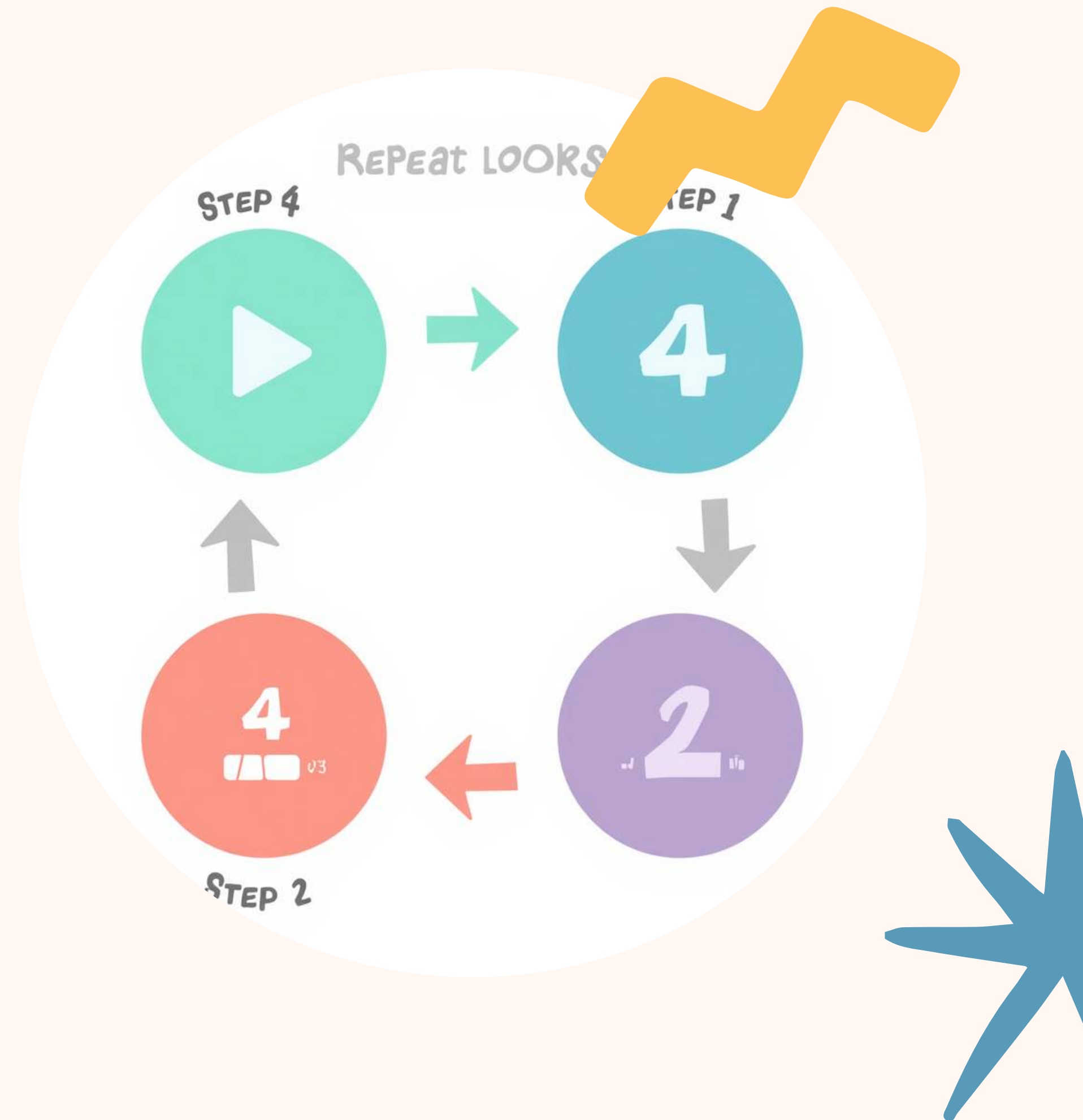
Step 1 → Run the blocks inside

Step 2 → Check: is the number done?

Step 3 → If NO, repeat again!

Step 4 → If YES, stop! ✓

Each time the loop runs, the computer counts down until it reaches zero!



DRAW A SQUARE!

THE SCRATCH CODE

To draw a square, we use the Repeat block 4 times. Inside it, we put: Move 100 steps, then Turn 90 degrees. The computer repeats this 4 times and draws a square!

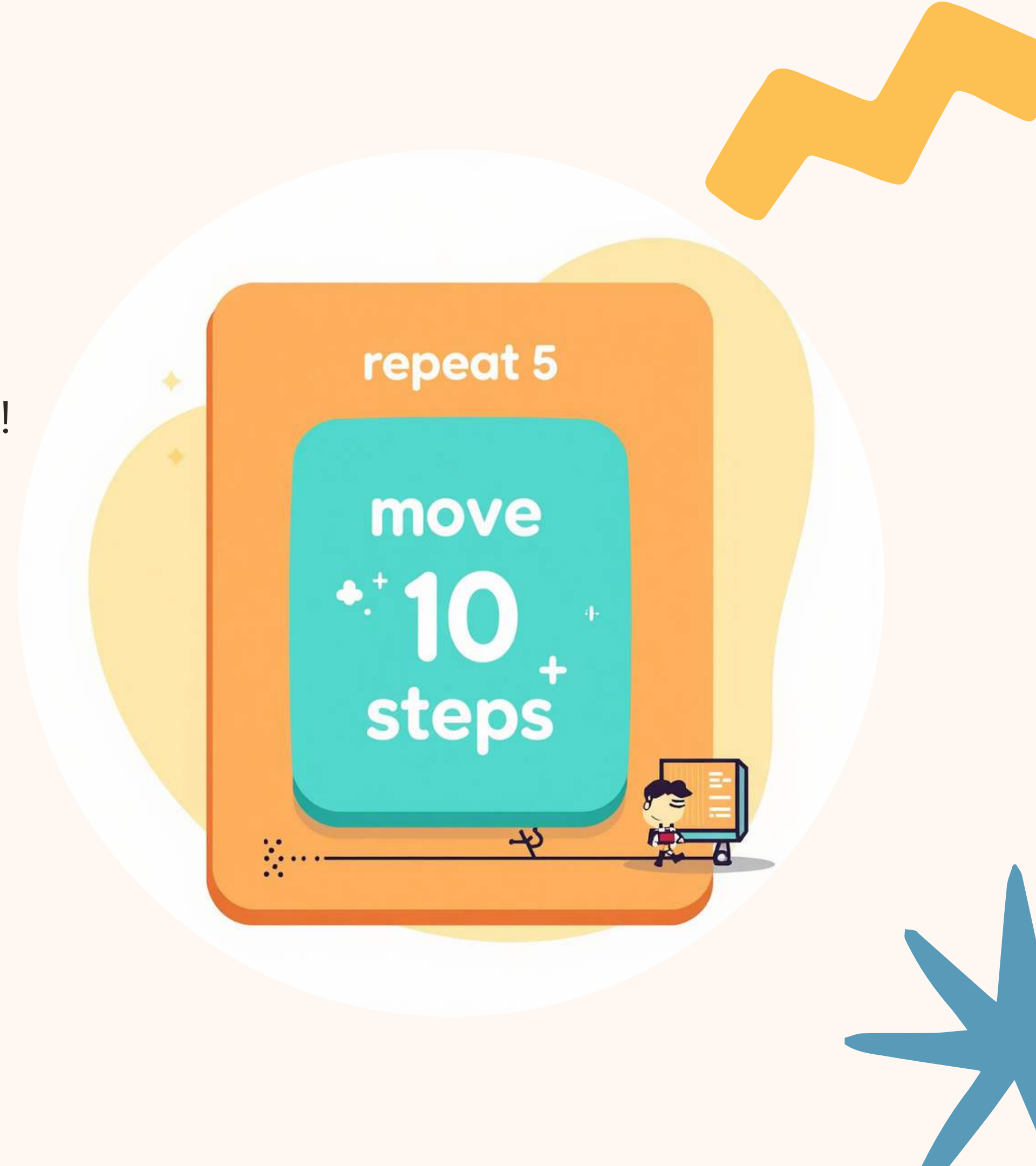
WHAT HAPPENS?

Each side of the square is 100 steps long. After 4 repeats, all 4 sides are drawn and you get a perfect square on the screen!



EXAMPLE 2: MOVE YOUR SPRITE

Make your sprite move forward 10 steps, 5 times!
Use the Repeat block with the number 5, and
put "Move 10 Steps" inside it.
The sprite will move forward 10 steps... again
and again... 5 times in a row!





**ACTIVITY 1:
YOUR TURN!
USE THE REPEAT BLOCK
TO MAKE YOUR SPRITE
JUMP UP AND DOWN 3
TIMES.
GIVE IT A TRY!**



COMMON MISTAKES WITH LOOPS

1. BLOCKS INSIDE REPEAT

Make sure your action blocks are placed inside the Repeat block. If they are outside, they won't repeat!

1. WRONG NUMBER OF REPEATS

Check the number in your Repeat block. If it's too big or too small, your sprite won't do what you want!

1. BLOCKS OUTSIDE REPEAT

Check that your blocks are snapped inside the Repeat block. If they are outside, the loop won't include them!



ACTIVITY 2: CREATE YOUR OWN LOOP!

Make a fun animation or movement using the Repeat block!

What will your sprite do?

🎨 Draw or imagine your idea:

- Will it spin? Jump? Dance?
- How many times will it repeat?
- Make it your own!

Be creative and have fun! 🌟



WHAT DID WE LEARN TODAY?



Coding tells computers what to do
Loops repeat actions to save time

The Repeat block repeats instructions
in Scratch
We used Repeat to draw and move
sprites



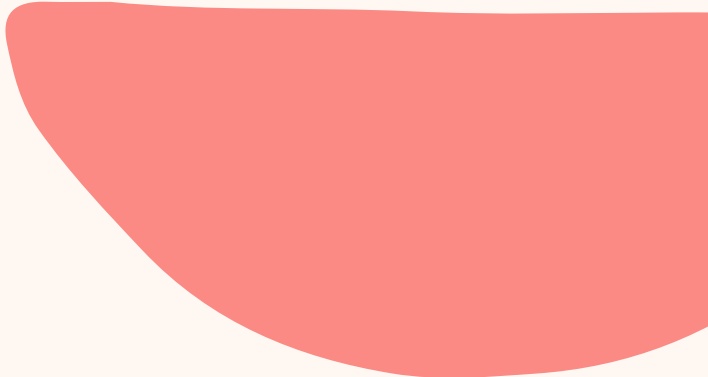
**DID YOU KNOW?
THE WORD "LOOP"
COMES FROM THE
IDEA OF GOING
AROUND AGAIN AND
AGAIN!**

KEEP CODING AND EXPLORING!

YOU ARE DOING AN AMAZING JOB!

Keep exploring, creating, and having fun with coding! Every great coder started just like you. What will you create next?





THANK YOU!

Any questions?

